

Task : Does the new page have a higher click-through rate than the old page ?

Terminology :

- Control group : the participants who use the current, unchanged environment .
- Treatment group : the participants who use the new version of the environment being tested .
- Conversion : the action we want the user to do .
same action applies to both groups

In Our Project :

Control group - ppl who see the old page
Treatment group - ppl who see the new page
Conversion - user clicked on the page .

The end decision is : should the treatment be applied ?
In our project : should the company launch the new page ?

Project Workflow:

- 1 - understand data ✓
- 2 - data quality & cleaning ✓
- 3 - data exploration ✓
- 4 - calculate conversion rates ✓
& measure the difference
- 5 - visualise results ✓
- 6 - hypotheses ✓
- 7 - statistical test ✓
- 8 - confidence interval ✓
- 9 - final recommendation ✓

Group	No. Users (n)	No. Conversions (x)	Conversion Rate (p)
Control	147202	17723	0.12039917935897611
Treatment	147276	17514	0.1189195795648985

Let p_c = true conversion proportion in control group

p_T = true conversion proportion in treatment group

Question : does the treatment group have a higher conversion proportion than the control group?

Hypotheses :

$$\begin{array}{l} H_0 : p_T = p_c = p \\ \quad (\text{i.e. } p_T - p_c = 0) \\ H_1 : p_T > p_c \\ \quad (\text{i.e. } p_T - p_c > 0) \end{array} \parallel \begin{array}{l} \text{choose a } 5\% \\ \text{significance level} \end{array}$$

Estimators :

we don't have p_c and p_T , but we can provide estimates, $\hat{p}_T$ and $\hat{p}_c$, from our sample.

$\hat{p}_T$ and $\hat{p}_c$ are unbiased estimators because ...

Test :

Under the assumption that H_0 is true, $p_T = p_c = p$

$$\Rightarrow \hat{p} \text{ can be calculated as a pooled proportion of } \hat{p}_T \text{ and } \hat{p}_c : \frac{x_T + x_c}{n_T + n_c} = \frac{17723 + 17514}{147202 + 147276}$$

$$p = 0.1196591936$$

Standard error

$$SE = \sqrt{\hat{p}(1-\hat{p})\left(\frac{1}{n_T} + \frac{1}{n_C}\right)} = 0.001196$$

(this is the standard deviation of the estimator $\hat{p}_T - \hat{p}_C$)

Standardising result (z-score)

$$Z = \frac{\hat{p}_T - \hat{p}_C}{SE} = \frac{-0.1189 - 0.1203...}{0.001196}$$

$$\approx -1.2369...$$

Critical Value

$$Z \sim N(0, 1^2) \Rightarrow P(Z > \alpha) = 0.05$$
$$\Rightarrow \alpha = 1.6448...$$

Conclusion

our test statistic $-1.23...$ does not exceed the critical value of $1.64...$

so insufficient evidence to reject H_0

insufficient evidence that the new landing page has a higher click-through rate than the old one.

p-value

$$P(Z > -1.2369...) = 0.8919...$$

The p-value represent how strong the evidence is against H_0 , in a quantifiable way. ... (more in notes)

Confidence interval

$$P(a < Z < b) = 0.95 \Rightarrow a = -1.95996398 \\ b \Rightarrow 1.95996398$$

$$P(-1.9599... < \frac{(P_T - P_C) - (\hat{P}_T - \hat{P}_C)}{SE} < 1.9599) = 0.05$$

$$\therefore P(-0.00148 + (-1.9599 \times SE) < P_T - P_C < -0.00148 + (1.9599 \times SE)) = 0.05$$

because calculating the CI does not assume P_T and P_C are equal, we use an unpooled standard error:

$$SE = \sqrt{\frac{\hat{P}_T(1 - \hat{P}_T)}{n_T} + \frac{\hat{P}_C(1 - \hat{P}_C)}{n_C}} = 1.19619364...$$

$$\therefore P(-0.00382... < P_T - P_C < 0.000864...) = 0.05$$

$$CI \text{ is } (-0.382\%, -0.0086\%)$$

This interval contains 0, supporting our hypothesis test's conclusion that the difference in the two groups' conversion proportions are consistent with random sampling variation.